

# FFGFT — Epistemological Position

## *4D Description, Natural Units, and Self-Referentiality*

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This document summarises the epistemological position of FFGFT. It addresses: (1) the 4D description (fractal, open, recursive), (2) equivalence of all descriptions in natural units, (3) structure of the vacuum and T-field, (4) FFGFT position on black holes, (5) self-referentiality of any inside description, and (6) why  $T^4$  as a posit is operationally superior to “pure information”. A closing remark draws the parallel to the Logos concept.

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## 1. The 4D Description Is Fractal, Open, and Recursive

FFGFT works with the 4D identification torus structure  $T^4$  — not a closed, deterministic system.

**Fractal:**  $D_f = 2.999867$  — subtle but physically consequential.

**Open:** Why  $T^4$ , why  $\xi = 1.333 \times 10^{-4}$  — remains explicitly open. No ontological completeness claim.

**Recursive:**  $\xi$ -ladder with natural cut-off  $N \approx 8-9$ , where  $Z_3^3$  topology and Fibonacci convergence coincide.

## 2. The 4D Description Does Not Apply to the Totality

FFGFT does *not* claim that the entire domain of existence is described by 4D. Derivation framework: *given  $T^4$  and  $\xi$ , everything else follows*. What lies before  $T^4$  remains open.

## 3. The Lower Bound: 4D Applies to the Absolutely Smallest

$$L_0 = \xi \cdot \square_P \approx 2.2 \times 10^{-39} \text{ m}$$

Below  $L_0$ ,  $T^4$  has no well-defined degrees of freedom. Boundary applies *downward*. Torus size is an emergent coherence length, not a fixed parameter.

## 4. Natural Units: No Privileged Description

$$E \leftrightarrow m \leftrightarrow 1/t \leftrightarrow f$$

$E = mc^2$ ,  $E = hf$ ,  $T \cdot m = 1$  ( $T$  = T-field operator, FFGFT's foundational relation),  $f = 1/t$ . None is privileged — all are projections of the same geometric reality on  $T^4$ . [Pascher 2026]

## 5. Consequence: Information vs. Geometry

“Information” as a description is a valid projection — but not more fundamental than the geometric description.

**Information is distinction.** Shannon (1948), Bateson (1972), IPI 2026: without distinction, there is no information. “Pure unrestrained information” without structure is indistinguishable from nothing — not a primitive, but the absence of everything.

**Consequence for Atticus's Postulate:** Continuous spin, unbounded velocity, no quantum numbers — this is not “pure information” but **the absence of all structure through which information would be defined**. Not connectable to any known or coherently conceivable physics.

## 6. The Vacuum Has Structure — No Field Without Structure

### 6.1. The Vacuum Is Not Empty

$\xi$ -field ground state: vacuum energy density as geometric property,  $T \cdot m = 1$  active in the ground state,  $L_0 = \xi \cdot \ell_P$  as structural lower bound.

### 6.1.1 Thermodynamic Equilibrium Changes Nothing

Even in total equilibrium, the field remains structured: field equations hold,  $T^4$  topology exists,  $\xi$  retains its value.

**Equilibrium is not the absence of structure.** Without field structure there is no concept of equilibrium.

## 6.2. FFGFT Is a Field Theory — No Field Without Structure

**No field without structure** — mathematical tautology. “Pure unstructured information” cannot carry a field, build a vacuum, or satisfy a field equation.

## 7. Black Holes — Horizon, Singularity, and Fundamental Information

### 7.1. Black Holes Are Points in the Universe, Not the Universe

Event horizon and cosmological horizon are **epistemic** boundaries — not ontological statements. We can only describe what we see.

### 7.2. Black Holes Are Not Singular in FFGFT

$L_0 = \xi \cdot \ell_P$  is an absolute lower bound — infinite density is structurally excluded.  $T \cdot m = 1$  remains valid. Black holes are  $T^4$  winding configurations. [Pascher 2026]

### 7.3. FFGFT Does Not Lose Fundamental Information

Winding numbers ( $n_\theta$ ,  $n_\phi$ ,  $n_3$ ,  $n_4$ ) are *topological invariants*. Information loss is an **access problem** (epistemic), not an **existence problem** (ontological). [Bekenstein 1973; Hawking 1974]

## 8. Self-Referentiality — We Always Describe From the Inside

### 8.1. We Are Part of the System

Neurons, electrochemical signals, conscious processes — all within  $T^4$ . The description is a process *within* the system acting as if it stands above it.

**We are part of the system. We always describe from the inside, never from the outside.**

### 8.2. The Thought Experiment of an Outside View

The attempt to adopt an outside view is itself a material process within  $T^4$ . Every apparent outside view is an inside view dressed as an outside view.

### 8.3. The Recursive Situation

Matter → Description → Matter → Description → ...

In FFGFT, this recursion does not terminate — consistent with its fractal, open, recursive character.

**Gödel Analogy [1931]:** A sufficiently powerful formal system cannot prove its own consistency from its own axioms. “Why  $T^4$ ?” remains explicitly open — structural honesty, not a gap.

**Why  $T^4$  and Not “Pure Information”?**  $T^4$  is **operational**: derivations (winding numbers,  $\xi$ ,  $L_0$ , field equations), measurable predictions. “Pure information” is **not operational**: no derivation, no distinction possible, coherence already presupposes structure.  $T^4$  proves itself. “Pure information” does not.

#### 8.4. Consequence for the Postulate of Pure Information

“Pure information before geometry” implicitly assumes an outside view — a standpoint beyond all structure. But this standpoint is itself a material process within  $T^4$ . FFGFT accepts this limit.

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### 9. The Limit of the First Ground: Philosophical and Theological Parallels

The question “Why  $T^4$ ?” and more fundamentally “Why  $\xi = 1.333 \times 10^{-4}$ ?” is not a weakness of FFGFT. It is the inevitable structure of any honest theory. The history of philosophy and theology has reached the same limit from different directions.

#### 9.1. Leibniz: Why Is There Something Rather Than Nothing?

**Gottfried Wilhelm Leibniz** posed the deepest metaphysical question in 1714: “Pourquoi il y a quelque chose plutôt que rien?” [Leibniz 1714] The principle of sufficient reason: everything has a reason, but the reason for the whole cannot come from within the whole. Parallel to FFGFT: “Why  $\xi$  and not some other value?” has the same structure — one level more concrete, and therefore more honestly formulable.

#### 9.2. Spinoza: No Outside — But Also No Perspective

**Baruch de Spinoza** resolved the problem radically: *Deus sive Natura* — God or Nature. [Spinoza 1677] No outside, because God is identical with Nature. But at a price: no perspective. “Why this Nature and not another?” is not answerable. FFGFT shares Spinoza’s result: no well-formed outside to  $T^4$ . But FFGFT claims less — not that  $T^4$  is everything, only that  $T^4$  is the framework within which we describe.

#### 9.3. Kant: The Thing-in-Itself Is Unknowable

**Immanuel Kant** drew a sharper boundary in 1781: between the world as we can know it (phenomena) and the world as it is in itself (noumena, thing-in-itself). [Kant 1781] Unknowable in principle — not from lack of intelligence, but because cognition is always mediated through the forms of intuition. “Why T<sup>4</sup>?” is the FFGFT version of Kant’s thing-in-itself.

#### 9.4. Gödel: No System Proves Its Own Completeness

**Kurt Gödel’s** incompleteness theorem 1931: in every sufficiently powerful formal system, there are statements neither provable nor refutable within the system. [Gödel 1931] “Why  $\xi$ ?” is not provable within FFGFT — the answer lies outside the system. Not a flaw but Gödel’s result applied to physical primitives. FFGFT names this explicitly.

#### 9.5. Wittgenstein: Whereof One Cannot Speak

**Ludwig Wittgenstein** closed the Tractatus in 1921 with: “Whereof one cannot speak, thereof one must be silent.” [Wittgenstein 1921] FFGFT makes the same decision: “Why T<sup>4</sup>?” lies outside the describable. Leaving explicitly open what one cannot say is Wittgenstein’s silence — actively formulated, not passively avoided.

#### 9.6. Panentheism and Tzimtzum: The Contradiction of Inside and Outside

**Alfred North Whitehead** (process theology, panentheism) [Whitehead 1929]: God is both in everything (immanent) and beyond (transcendent). Both together is a contradiction — a system cannot fully encompass its own interior and simultaneously stand outside it. Gödel proved this formally; theologians call it “mystery”. The **Tzimtzum** (Isaac Luria, 16th c.) [Scholem 1941]: God withdraws to make room for creation — mythological formulation of the same geometric problem.

#### 9.7. The Creator God Outside Time: The First Contradiction

A being outside time knows no before and no after. But “initiate” presupposes precisely that — a temporal act. **Augustine** felt this: “Quid faciebat Deus antequam faceret caelum et terram?” [Augustine 397–400] His answer (God created time with creation) merely relocates the problem. Moreover: a God who does not know in detail what will emerge is not omniscient — loses his status as *necessarium ens*. FFGFT is structurally the same as such a creator God who posits a single primitive ( $\xi$ ) and lets the consequences unfold — without the theological contradictions: no timelessness that simultaneously acts, no claim to omniscience.

**The common limit:** All these traditions — Leibniz, Spinoza, Kant, Gödel, Wittgenstein, Whitehead, Luria, Augustine — reach the same limit: the first ground cannot ground itself from within itself. FFGFT reaches the same limit

from the direction of physics. The difference: FFGFT names it explicitly, rather than bypassing it through inspiration, mystery, or silent positing.

## Closing Remark: “In the Beginning Was the Word”

The discussion about the primitive leads, in principle, consciously or unconsciously, to: “In the beginning was the Word” (John 1:1). The Logos concept is philosophically precise: the Word is structure, order, distinction — exactly what information in the Shannon sense presupposes.

The postulate of “pure, unrestrained information” and the Johannine Logos share the same structural problem: both require someone *outside the system* to speak. The writers of the Bible were human — material beings within the system.

**The difference in one sentence:** The Bible claims an outside view it cannot have. FFGFT relinquishes this claim and names the boundary explicitly.

## Summary

| Claim                                            | FFGFT Position                                               |
|--------------------------------------------------|--------------------------------------------------------------|
| 4D is the only possible description              | No — 4D is the framework, ontologically open                 |
| 4D applies to the totality of existence          | No — FFGFT makes no total claim                              |
| 4D applies to the absolutely smallest            | Yes — $L_0 = \xi \cdot l_P$ sets the lower bound             |
| Below $L_0$ nothing more exists (in FFGFT)       | Yes — no stable degrees of freedom                           |
| Geometry is privileged over energy / mass / time | No — all projections are equivalent                          |
| Unstructured information as primitive            | Contradiction — information is distinction, hence structure  |
| Atticus’s postulate is physically connectable    | No — empty name, not operational, no field equation possible |
| The vacuum is structureless                      | No — $\xi$ -field ground state, geometrically structured     |
| A field can exist without structure              | No — mathematical tautology                                  |
| Black holes are singular in FFGFT                | No — $L_0$ structurally prevents infinite density            |
| Black holes destroy information                  | No — topological invariants are preserved                    |

| Claim                                              | FFGFT Position                                                        |
|----------------------------------------------------|-----------------------------------------------------------------------|
| Horizons are ontological boundaries                | No — epistemic limits of the observable                               |
| We describe FFGFT from outside                     | No — material $T^4$ processes describing from within                  |
| The recursion matter→description terminates        | No — open recursion, consistent with FFGFT's character                |
| $T^4$ and “pure information” are equivalent posits | No — $T^4$ is operational; “pure information” has no derivation power |
| “Why $T^4$ ?” is answered                          | No — structural honesty, not a gap                                    |
| The Bible adopts an outside view                   | Claim from inside — FFGFT names the boundary explicitly               |

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